Status	Finished
Started	Wednesday, 21 October 2026, 12:20 PM
Completed	Wednesday, 21 October 2026, 12:26 PM
Duration	6 mins 46 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It requires the list to be sorted ✓
c. It does not work with negative numbers
d. It can only be applied to large lists

Question **2**

Incorrect

Mark 0.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 94 and 99
b. 89 and 94
c. 90 and 99
d. 90 and 100 ✗

Question **3**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Searching ✓
b. Mining
c. Finding
d. Discovering

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Dynamic search
- b. Linear search ✓
- c. Binary search
- d. Hashing

Question **5**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is large and sorted
- b. When the list is small and unsorted ✓
- c. When the list is large and unsorted
- d. When the list is small and sorted

Question **6**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

- a. When the item is somewhere in the middle of the array ✓
- b. When the item is the last element in the array
- c. Item is the last element in the array or item is not there at all
- d. When the item is not the array at all

Question **7**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Both (1) & (3)

b. Linear search ✓

c. Hash search

d. Binary search

Question **8**

Incorrect

Mark 0.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts

b. Compare the target element with the first element in the list

c. Compare the target element with the middle element in the list

d. Sort the list ✗

Question **9**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list

b. Binary search has a lower time complexity on large, sorted lists ✓

c. Binary search can find multiple instances of the target element

d. Binary search works on unsorted lists

Question **10**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By dividing the list into halves
- c. By sorting the list first
- d. By using a hash function

Question **11**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

- a. n ✓
- b. log n
- c. n-1
- d. n/2

Question **12**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It works on unsorted lists
- b. It compares elements sequentially
- c. It always starts from the beginning of the list
- d. It can be applied only if the list is sorted ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the list is sorted
- b. When the list contains duplicate elements
- c. When the search algorithm finishes
- d. When the element is found in the list ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By comparing each element sequentially
c. By dividing the list length by two ✓
d. By starting from the first element

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. At the beginning of the list
c. In the left sublist ✓
d. In the entire list